EECE5554 — Robotics Sensing and Navigation

Lab 3: IMU Noise and Performance Characterization

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1 Objective

The purpose of this lab is to understand sources of noise in inertial measurement units (IMUs) and in the surrounding environment, and to develop the skills necessary to characterize and select IMU sensors for different robotic applications. Specifically, the lab involves writing a ROS2 device driver for the VectorNav VN-100 IMU, collecting stationary and dynamic motion data, and performing Allan variance analysis on long-duration datasets to extract key noise parameters.

2 Methodology

2.1 Hardware

The sensor used in this lab is the VectorNav VN-100 IMU, a miniature, high-performance inertial measurement unit capable of providing accelerometer, gyroscope, magnetometer, and orientation data. The sensor communicates over USB serial at 115200 baud and outputs data in the `$VNYMR` NMEA-style sentence format.

2.2 ROS2 Device Driver

A ROS2 device driver was written in Python to interface with the VN-100 over USB serial. The driver performs the following key functions:

- Opens the serial port at 115200 baud with the port specified at runtime
- Writes the VectorNav register command `$VNWRG,07,40` to configure the sensor output rate to 40 Hz

- Reads and parses `$VNYMR` sentences, extracting yaw, pitch, roll, magnetometer, accelerometer, and gyroscope data
- Validates each sentence using XOR checksum verification
- Publishes data on the `/imu` topic using a custom `imu_msgs/msg/IMUmsg` message type

The custom message `IMUmsg.msg` contains a `std_msgs/Header`, a `sensor_msgs/Imu`, a `sensor_msgs/Magnet` and the raw `$VNYMR` string. Euler angles from the sensor are converted to quaternions analytically using the standard roll-pitch-yaw to quaternion formula. The driver was verified to publish at approximately 40 Hz on both simulated and real hardware.

2.3 Ubuntu USB Latency Configuration

Standard Ubuntu has a default USB serial latency of 16 ms, which causes issues with high-rate sensors. Two udev rules were configured to address this:

- `50-VN-100.rules` — assigns the correct permissions to the VN-100 device based on its vendor ID (`0403`) and product ID (`6001`)
- `49-USB-LATENCY.rules` — sets the FTDI serial latency timer to 1 ms

The latency timer was verified before each data collection session with:

```
cat /sys/bus/usb-serial/devices/ttyUSB0/latency_timer
```

2.4 Data Collection

Three categories of data were collected:

1. **Stationary data** — each team member individually collected 5–10 minutes of stationary IMU data with the sensor placed on a stable surface away from computers and moving objects.
2. **Motion data** — one dataset per team was collected with a teammate moving the IMU, one managing the rosbag, and one recording a video of the motion. The IMU was moved along three distinct 2D paths traced on a whiteboard: a square, a sine wave, and a circle, each with slight variation in depth to introduce a controlled 3D component.
3. **Long-duration data** — pre-recorded bag files provided via Piazza were used for Allan variance analysis. The dataset used in this report is `5_hour_imu_data`, a 6.6-hour recording at approximately 40 Hz containing 914,712 messages.

All bags were recorded using `ros2 bag record` at 40 Hz in `db3` format.

2.5 Data Analysis

Analysis was performed using Python scripts in the `Analysis/` directory. Time series plots and orientation histograms were generated using `matplotlib`. Allan variance analysis was

performed using the `allantools` library with the overlapping Allan deviation (`oadev`) function using `taus='all'` and `data_type='freq'`.

3 Experiments and Results

3.1 Stationary Data Analysis

Stationary data was collected for approximately 7 minutes (416 seconds, 16,669 messages) with the VN-100 placed on a stable surface. The IMU was positioned away from the computer and other sources of electromagnetic interference.

3.1.1 Gyroscope Time Series

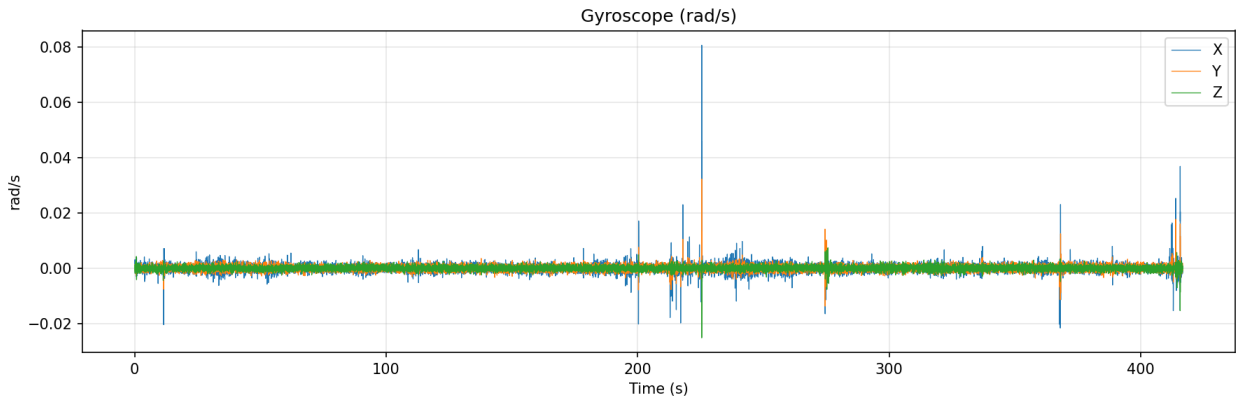


Figure 1: Gyroscope angular velocity (X, Y, Z axes) over the stationary recording period. All axes remain near zero as expected for a stationary sensor, with occasional spikes due to environmental vibration.

3.1.2 Accelerometer Time Series

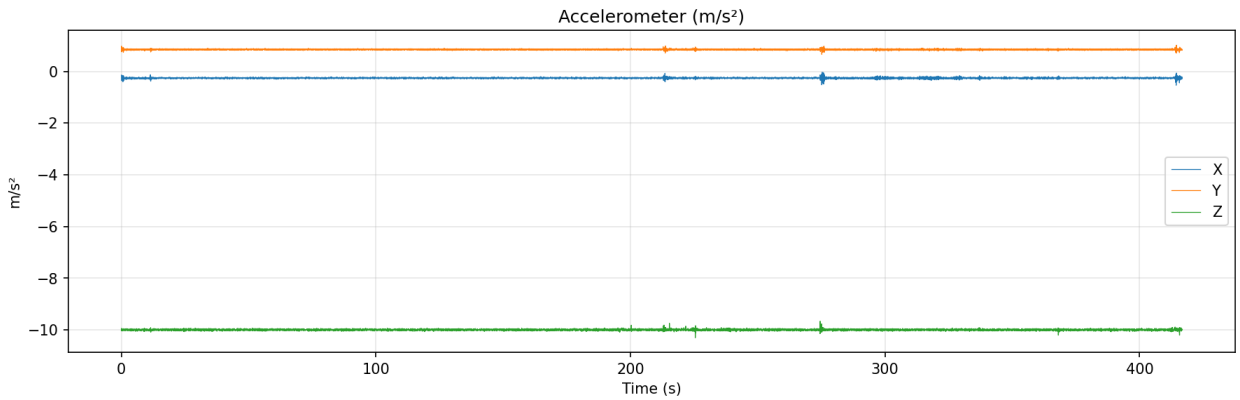


Figure 2: Accelerometer readings (X, Y, Z axes) over the stationary recording period. The Z axis reads approximately -9.8 m/s^2 due to gravity. Occasional disturbance spikes are visible around 200–270 s, likely caused by environmental vibration.

3.1.3 Magnetometer Time Series

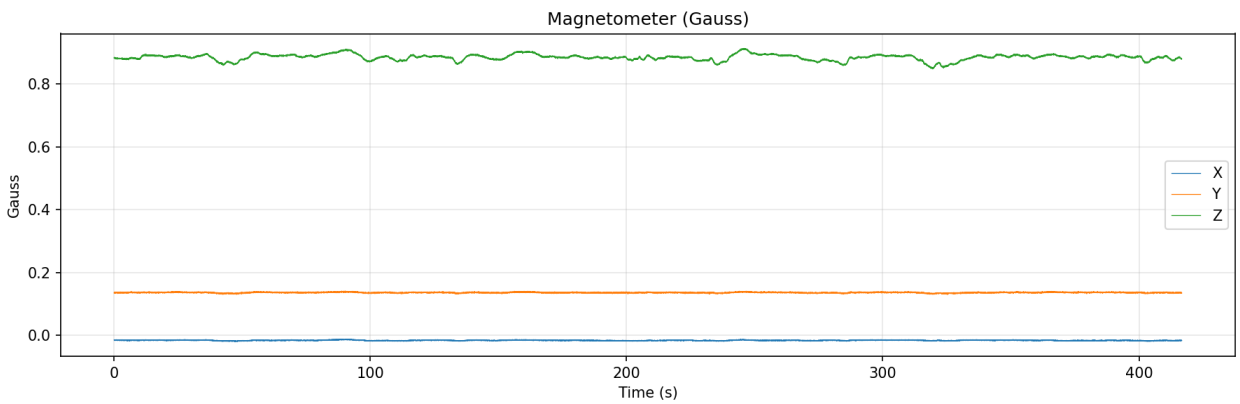


Figure 3: Magnetometer readings (X, Y, Z axes) over the stationary recording period. The Z axis dominates at approximately 0.87 Gauss, consistent with the local magnetic field orientation at the measurement location.

3.1.4 Orientation Time Series

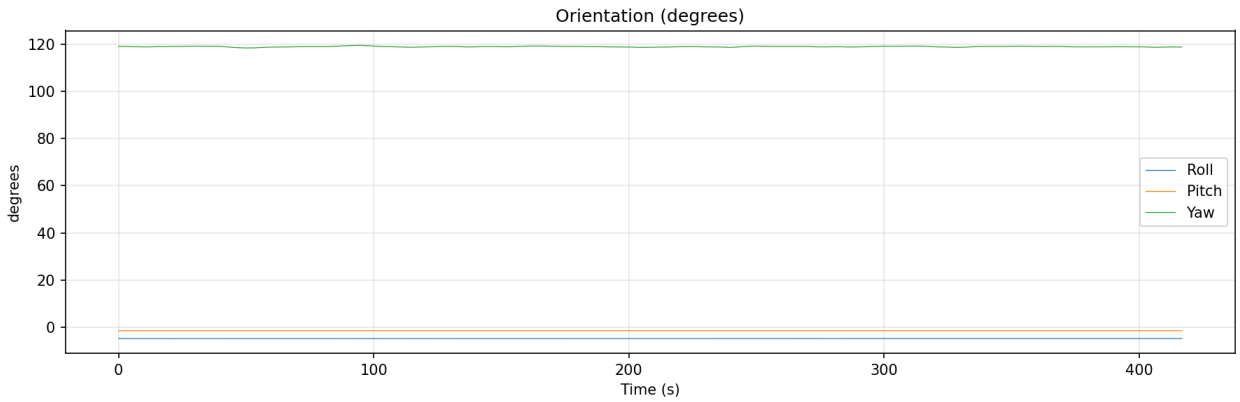


Figure 4: Euler angle orientation (Roll, Pitch, Yaw) over the stationary recording period. Yaw is approximately 119, Roll approximately -4.8, and Pitch approximately -1.5, all stable throughout the recording.

3.1.5 Orientation Histograms and Statistics

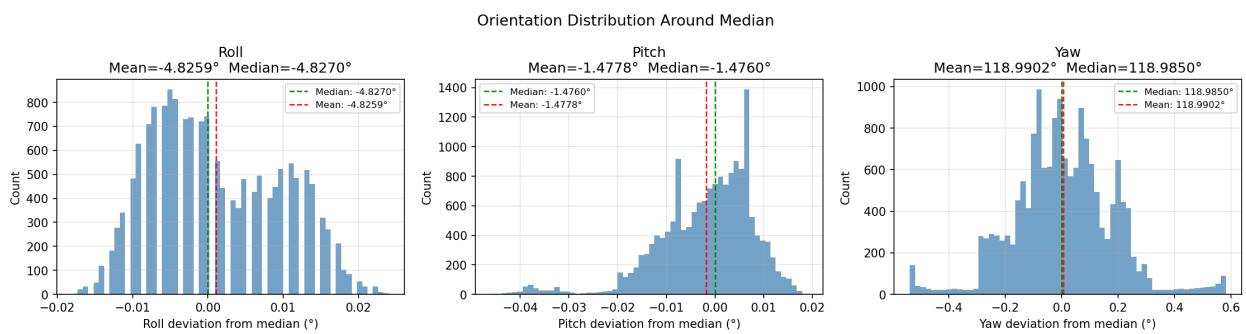


Figure 5: Histogram of orientation deviation from median for Roll, Pitch, and Yaw. Roll and Pitch distributions closely approximate a Gaussian. Yaw shows a broader, less symmetric distribution reflecting greater sensitivity to magnetic interference.

Table 1 summarizes the mean, median, and standard deviation of the orientation time series data.

Table 1: Orientation statistics from stationary data.

Axis	**Mean (°)**	**Median (°)**	**Std Dev (°)**
Roll	-4.8259	-4.8270	0.0085
Pitch	-1.4778	-1.4760	0.0098
Yaw	118.9902	118.9850	0.1771

3.2 Motion Data Analysis

Motion data was collected as a group over approximately 5 minutes (313 seconds, 12,524 messages). One teammate moved the IMU through three distinct 2D paths traced on a whiteboard — a square, a sine wave, and a circle — each with slight variation in depth to introduce a controlled 3D component. A second teammate managed the rosbag recording and a third recorded a video of the motion for later alignment.

3.2.1 Motion Overview

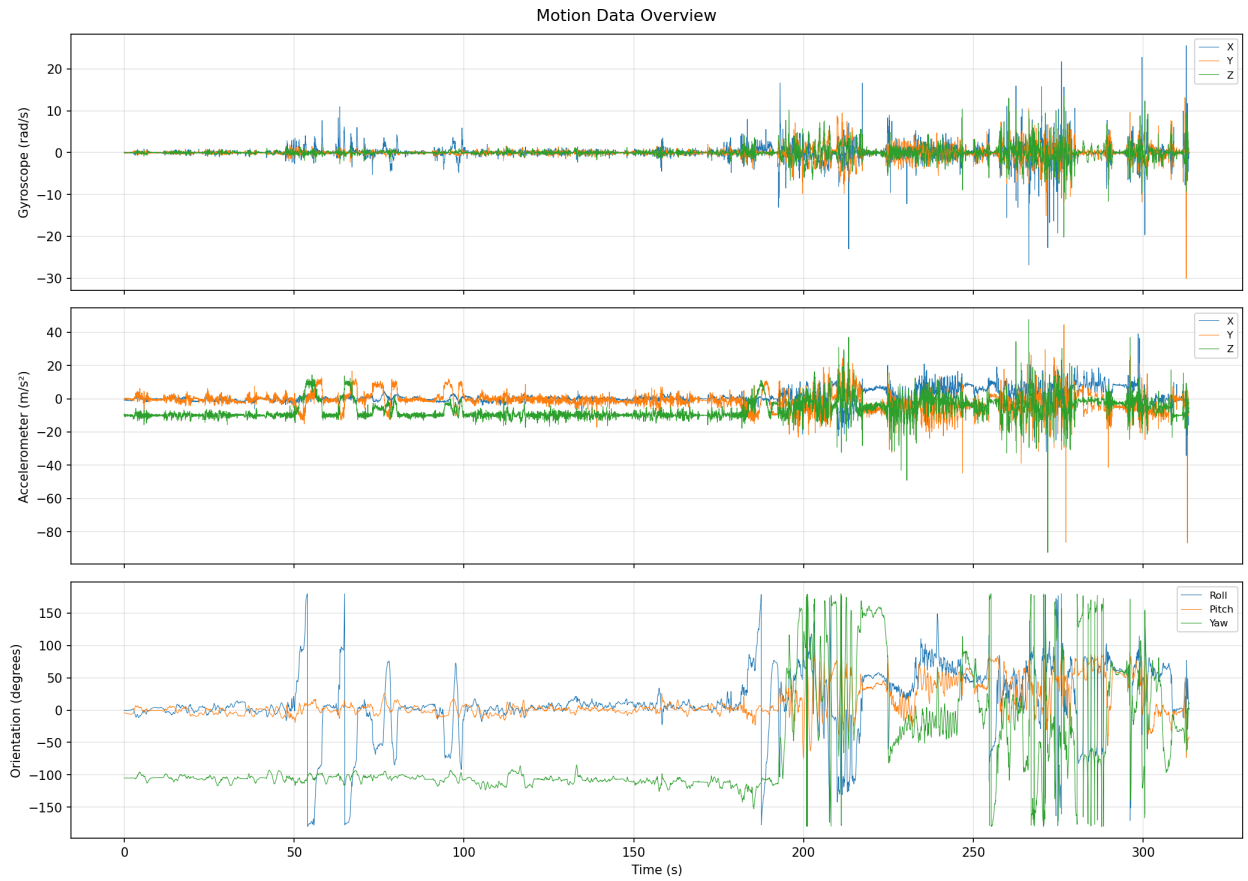
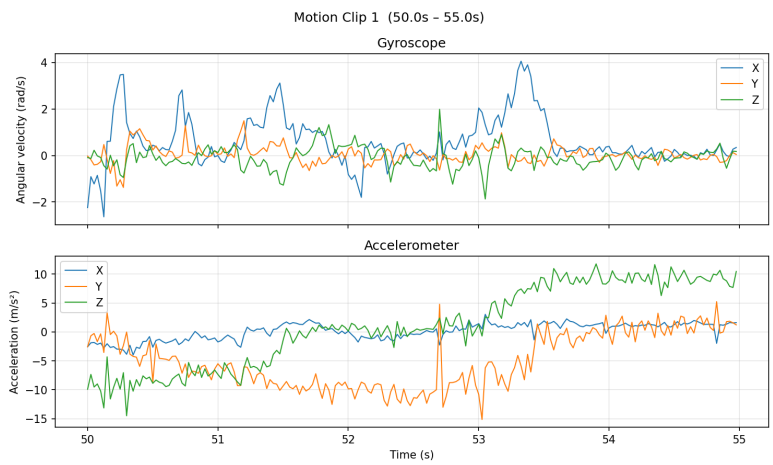
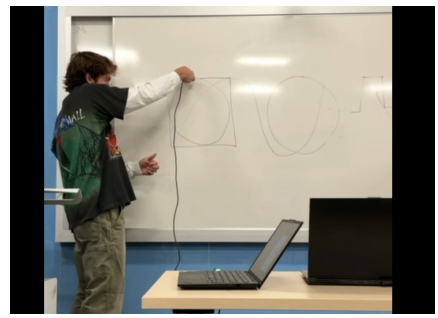


Figure 6: Overview of the full motion dataset showing gyroscope angular velocity, accelerometer readings, and Euler angle orientation across the entire 313-second recording. Three distinct motion phases are visible: initial single-axis rotation (0–100 s), moderate multi-axis motion (100–190 s), and high-dynamic complex motion (190–313 s).

3.2.2 Motion Clip 1 — Square Path (50–55 s)



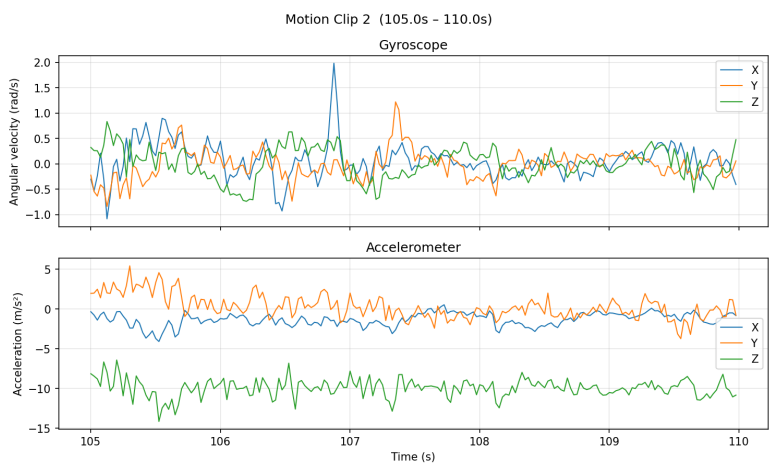
(a) IMU data for clip 1 (50–55 s).



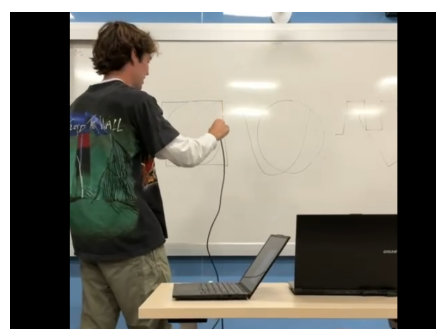
(b) Video screengrab at clip 1.

Figure 7: Motion clip 1 — tracing a square path on the whiteboard. Gyroscope X dominates with peaks up to ±4 rad/s, indicating rotation primarily about the X axis. The accelerometer Z axis shifts from −10 to +10 m/s^2 as the IMU is rolled through the corners of the square. The slight depth variation introduces a small response in all three accelerometer axes.

3.2.3 Motion Clip 2 — Sine Wave Path (105–110 s)



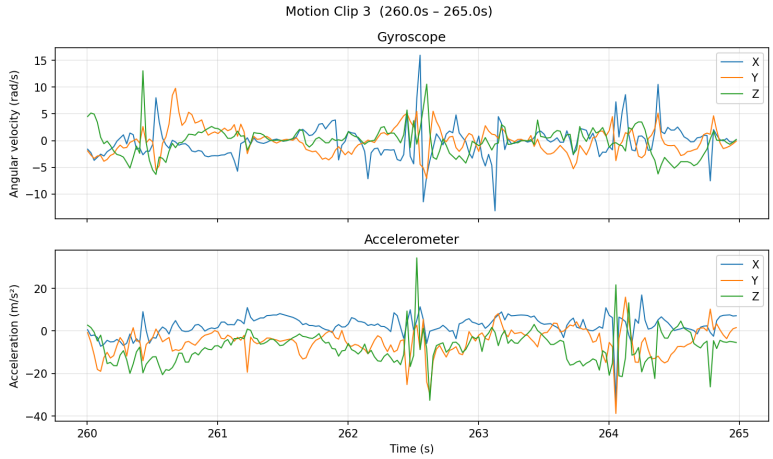
(a) IMU data for clip 2 (105–110 s).



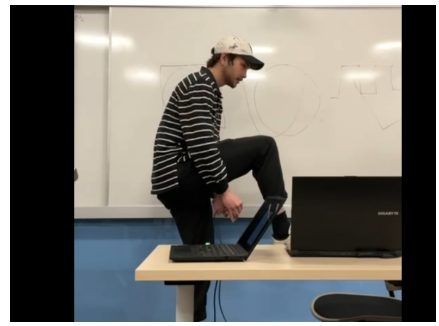
(b) Video screengrab at clip 2.

Figure 8: Motion clip 2 — tracing a sine wave path on the whiteboard. All three gyroscope axes show moderate, sustained activity at ±1 rad/s, consistent with the smooth, oscillatory nature of a sine wave. The accelerometer Z axis remains near −10 m/s^2, indicating the IMU stayed mostly upright while being swept through the wave. The periodic reversal of gyro X and Y reflects the alternating curvature of the sine path.

3.2.4 Motion Clip 3 — Random Motion (260–265 s)



(a) IMU data for clip 3 (260–265 s).



(b) Video screengrab at clip 3.

Figure 9: Motion clip 3 — completely random, unconstrained motion performed during the final minute of the recording. All gyroscope axes are active with peaks up to ±10 rad/s and accelerometer readings span ±40 m/s^2 across all axes. The high dynamic range and simultaneous response across all axes reflects the unconstrained nature of the motion, with no dominant axis or predictable pattern.

3.3 Allan Variance Analysis

Allan variance analysis was performed on the `5_hour_imu_data` bag file, a 6.6-hour stationary dataset recorded at approximately 40 Hz (914,712 messages). The overlapping Allan deviation (`oadev`) was computed for gyroscope and accelerometer axes. Three noise parameters were extracted:

- **N** — Angle Random Walk (ARW), read at $\tau = 1$ s
- **B** — Bias Instability, at the local minimum of the Allan deviation
- **K** — Rate Random Walk (RRW), estimated from the slope after the minimum

3.3.1 Gyroscope Allan Variance

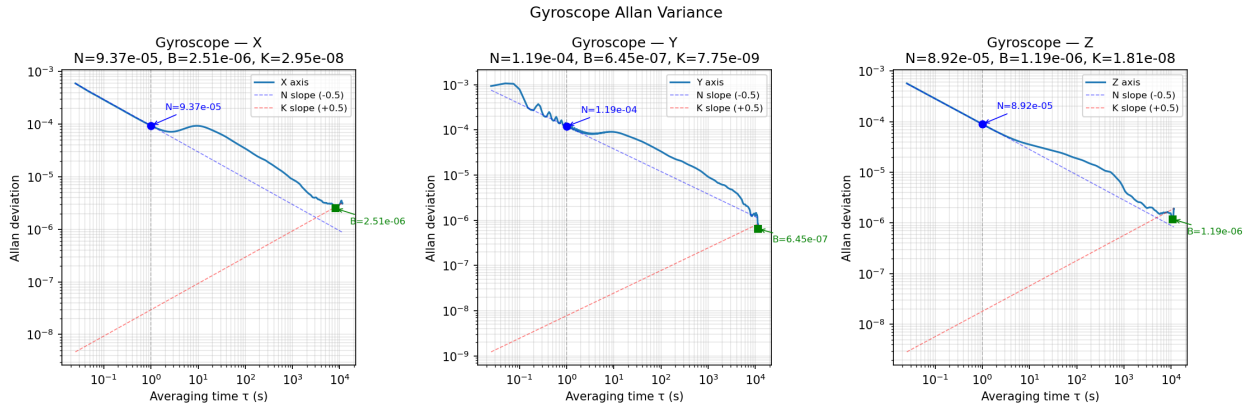


Figure 10: Allan deviation plots for gyroscope X, Y, and Z axes. The blue dashed line shows the $\tau^{-0.5}$ reference slope (angle random walk region) and the red dashed line shows the $\tau^{+0.5}$ reference slope (rate random walk region). N is marked at $\tau = 1$ s and B at the local minimum. Notable bumps in the Gyro Y curve suggest periodic environmental disturbance during the recording.

3.3.2 Accelerometer Allan Variance

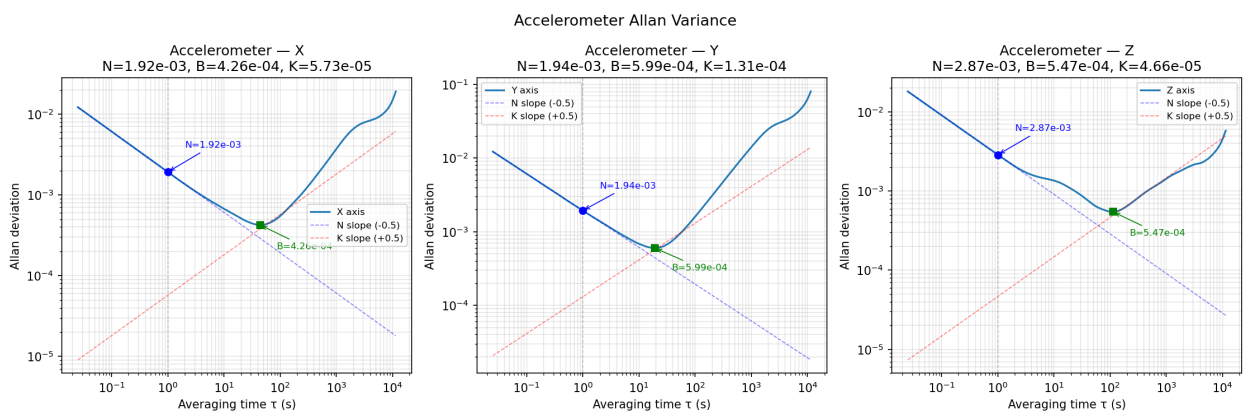


Figure 11: Allan deviation plots for accelerometer X, Y, and Z axes with the same reference slope annotations as the gyroscope plots. All three axes show a clear noise floor minimum followed by rising noise at longer averaging times.

3.3.3 Extracted Parameters

Table 2 summarizes the extracted noise parameters for all axes.

Table 2: Allan variance noise parameters extracted from the 6.6-hour dataset.

Sensor	**Axis**	**N (ARW)**	**B (Bias)**	**K (RRW)**
Gyro	X	9.37×10^{-5} rad/s/$\sqrt{\text{s}}$	2.51×10^{-6} rad/s	2.95×10^{-8} rad/s$\cdot\sqrt{\text{s}}$
Gyro	Y	1.19×10^{-4} rad/s/$\sqrt{\text{s}}$	6.45×10^{-7} rad/s	7.75×10^{-9} rad/s$\cdot\sqrt{\text{s}}$
Gyro	Z	8.92×10^{-5} rad/s/$\sqrt{\text{s}}$	1.19×10^{-6} rad/s	1.81×10^{-8} rad/s$\cdot\sqrt{\text{s}}$
Accel	X	1.92×10^{-3} m/s^2/$\sqrt{\text{s}}$	4.26×10^{-4} m/s^2	5.73×10^{-5} m/s$^2\cdot\sqrt{\text{s}}$
Accel	Y	1.94×10^{-3} m/s^2/$\sqrt{\text{s}}$	5.99×10^{-4} m/s^2	1.31×10^{-4} m/s$^2\cdot\sqrt{\text{s}}$
Accel	Z	2.87×10^{-3} m/s^2/$\sqrt{\text{s}}$	5.47×10^{-4} m/s^2	4.66×10^{-5} m/s$^2\cdot\sqrt{\text{s}}$

4 Questions and Answers

Q1. What can you say about the noise in your short stationary data?

The stationary data shows that all three gyroscope axes remain very close to zero (within ± 0.005 rad/s) for the majority of the recording, consistent with a high-quality MEMS IMU at rest. The accelerometer Z axis holds steady at approximately -9.8 m/s^2 due to gravity, with X and Y near zero. The magnetometer is stable throughout, with Z dominating at ≈ 0.87 Gauss. Notable spikes are visible in both gyroscope and accelerometer data around 200–270 s, with gyro peaks reaching ± 0.08 rad/s and accel Z dipping below -10.4 m/s^2. These are likely caused by foot traffic or vibration transmitted through the table surface, since the IMU was placed in a shared workspace environment. The noise is otherwise low-amplitude and broadband, characteristic of thermal and quantization noise in the sensor electronics.

Q2. What was the median, mean, and distribution?

The orientation statistics are summarized in Table 1. The mean and median are very close for all three axes, indicating a symmetric noise distribution with no significant drift over the 7-minute recording. As shown in Figure 5, Roll and Pitch distributions closely approximate a Gaussian centered around their respective medians, with small standard deviations of 0.0085 and 0.0098 respectively. Yaw shows a noticeably broader distribution with a standard deviation of 0.1771 — approximately 20 times larger than Roll and Pitch. This is expected: yaw is determined by the magnetometer, which is more susceptible to local magnetic interference than the accelerometer-based roll and pitch estimates.

Q3. How do your measurements compare to the performance listed in the VN-100 datasheet?

The VN-100 datasheet specifies a gyroscope in-run bias of 5°/hr and an accelerometer in-run bias of less than 0.04 mg. Converting to consistent units: the gyro bias stability

specification corresponds to 2.42×10^{-5} rad/s, and the accelerometer bias stability specification corresponds to 3.92×10^{-4} m/s^2.

Comparing against our extracted values in Table 2, the gyroscope bias instability values of 6.45×10^{-7} to 2.51×10^{-6} rad/s are well within — and in fact better than — the datasheet specification, suggesting the gyroscope is performing at or above its rated level. The accelerometer bias instability values of 4.26×10^{-4} to 5.99×10^{-4} m/s^2 are slightly above the datasheet specification of 3.92×10^{-4} m/s^2, indicating the accelerometer performance is marginally worse than the rated specification under the conditions of this recording. The datasheet does not explicitly list angle random walk or rate random walk parameters, so a direct comparison for those metrics is not possible from the available documentation.

Q4. If the performance in your dataset was worse than the datasheet, what sources of noise do you think were present? Why?

The gyroscope bias instability values are better than the datasheet specification. However, the accelerometer bias instability is marginally worse than the rated 3.92×10^{-4} m/s^2. Evidence of environmental noise sources was observed in our own stationary data collection. The gyroscope and accelerometer time series (Figures 1 and 2) show clear spikes around 200–270 s, most likely caused by foot traffic or vibration transmitted through the table surface in the shared workspace where data was collected. The IMU was also placed in proximity to a laptop computer, which introduces electromagnetic interference that can couple into the sensor electronics. These same categories of disturbance — mechanical vibration and electromagnetic interference from nearby equipment — are the most likely contributors to the accelerometer bias instability exceeding the datasheet specification in the long-duration dataset, since the datasheet value is measured under controlled laboratory conditions free from such environmental noise.

Q5. How did you align the video and VectorNav action data?

Video and IMU data were aligned by starting both recordings simultaneously. A verbal audio cue was used as a countdown, after which the video recording and the `ros2 bag record` command were triggered at the same instant. This meant the rosbag timestamp at $t = 0$ corresponded directly to the first frame of the video, eliminating the need for any post-hoc synchronization or offset correction. Motion clips were then identified by inspecting the IMU time series in Figure 6 for distinctive features and reading the corresponding timestamps directly, with the matching video frames found at those same elapsed times.

Q6. Did the IMU data correlate well with video of the actual motion?

Yes, the IMU data correlated well with the observed motion in all three clips. In Clip

1 (Figure 7), the square path produced sharp transitions in gyro X as the IMU turned through the corners, consistent with the right-angle direction changes visible in the video. In Clip 2 (Figure 8), the sine wave produced a smooth, oscillatory response in the gyroscope with the periodic reversals matching the wave frequency observed in the video. In Clip 3 (Figure 9), the completely random motion performed in the final minute produced the most complex multi-axis response, with all gyro and accel axes active simultaneously and no discernible pattern — consistent with the unconstrained, unpredictable movements observed in the video. The primary limitation in correlation was the manual video alignment — a small timing offset of up to 1–2 seconds was possible, which could cause slight mismatches between individual features in the plots and the video frames.

Q7. Can you make any inferences about the conditions under which your long-duration data was taken?

The long-duration bag file was provided as a pre-recorded dataset and the exact conditions under which it was taken are not known. However, several inferences can be made from the Allan variance plots themselves (Figure 10). The Gyro Y axis shows notable irregular bumps in its Allan deviation curve at intermediate averaging times, which is characteristic of a periodic or quasi-periodic noise source. This suggests the presence of a machine operating nearby during the recording — such as building HVAC, an elevator, or other rotating machinery introducing vibration at a characteristic frequency. The Allan deviation curves for all gyro axes continue decreasing at very long averaging times ($\tau > 1000$ s) without reaching a clear flat minimum, suggesting the recording environment was not sufficiently quiet for the true bias instability floor to be revealed, or that the 6.6-hour duration was not long enough for the slowest noise components to average out. The accelerometer curves show a clearer minimum at moderate averaging times, indicating the accel measurement environment was relatively more stable than the gyro environment during the recording.

Q8. What do your results tell you about the conditions you should measure Allan variance under to understand the best sensor performance possible?

The results highlight several requirements for ideal Allan variance measurement conditions. First, the sensor should be placed on a vibration-isolated surface — such as a foam pad or optical table — to decouple building and equipment vibrations from the sensor. Second, the measurement environment should be electromagnetically quiet, away from computers, motors, power supplies, and HVAC systems, all of which introduce periodic noise visible as bumps in the Allan deviation curve. Third, the temperature should be stable throughout the recording, as thermal drift introduces slow bias changes that corrupt the long-averaging-time portion of the curve. Fourth, the recording duration should be as long as practically possible — ideally 10 or more hours — to allow the bias instability minimum to be clearly resolved. Finally, the sensor should be in its intended powered-on thermal steady state before recording begins, as warm-up drift can corrupt

the early part of the dataset.

Q9. If you were trying to characterize a new sensor with unknown performance, briefly explain what, where, and how you would measure to understand sources of noise in the sensor and environment, and the performance you could expect under its operating conditions.

A systematic three-phase characterization approach would be used. In the first phase, a short stationary recording of 5–10 minutes would be collected in the intended operating environment to establish the baseline noise floor, distribution type, and identify any obvious environmental interference. In the second phase, a long-duration recording of at least 6–10 hours would be collected in the quietest available environment — ideally on a vibration-isolated surface in a temperature-controlled room away from other equipment — to perform Allan variance analysis and extract the intrinsic sensor parameters N, B, and K. In the third phase, a second long-duration recording would be collected in the actual operating environment (e.g. on a robot, near motors and computers) to characterize how the environment degrades sensor performance relative to the intrinsic baseline. Comparing the two Allan variance curves would allow separation of sensor noise from environmental noise, and the operating-environment curve would give the realistic performance to expect during deployment.

5 Conclusion

This lab successfully demonstrated the full pipeline for IMU characterization using the VectorNav VN-100. A ROS2 device driver was implemented in Python, verified at 40 Hz on both simulated and real hardware, and used to collect stationary, motion, and long-duration datasets.

Stationary analysis showed that the VN-100 produces low-noise, stable measurements with Roll and Pitch standard deviations below 0.01 and Yaw standard deviation of 0.18. Occasional environmental disturbances were visible as spikes in the gyro and accel time series, highlighting the sensitivity of MEMS IMUs to vibration in shared workspace environments. Orientation distributions for Roll and Pitch were approximately Gaussian, consistent with white noise dominating at short timescales.

Motion data analysis confirmed that the IMU data correlated well with the observed whiteboard tracing motions. The three clips — square, sine wave, and random motion — each produced distinct and interpretable IMU signatures, demonstrating the sensor's ability to capture both simple and complex 3D motions.

Allan variance analysis of the 6.6-hour dataset extracted gyroscope ARW values of 8.9–11.9$\times 10^{-5}$ rad/s/$\sqrt{\text{s}}$ and accelerometer ARW values of 1.9–2.9 $\times 10^{-3}$ m/s^2/$\sqrt{\text{s}}$, consistent with a high-performance MEMS IMU. The gyroscope bias instability was found to be better than the datasheet specification, while the accelerometer bias instability was marginally above the rated value due to environmental noise during the long-duration recording. Anomalies

in the Gyro Y curve suggest periodic environmental disturbance, reinforcing the importance of controlled measurement conditions for accurate sensor characterization. Overall, the VN-100 demonstrates performance suitable for short-duration robotic navigation tasks, though accumulated bias drift limits its utility for long-duration dead-reckoning applications without external corrections.